

Mikhail Razakov

Compiler Tools Developer

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Moscow / remote
from September 2026
up to 20 h/week

In UniversalToolchain/Wist2, I built a typed API, structured diagnostics, a versioned trace, parity/negative tests, and package smoke checks. For the PS-form Analyzer, I built a generator and test harness with an exact oracle, metamorphic tests, and WA/TL hunting modes.

8 additional defects

found by a second verification pass and covered by regressions

1,325 tests

parity, negative, lifecycle, and architecture checks

Exact oracle + metamorphic tests

case generation, WA/TL hunting, and counterexample retention

EXPERIENCE

2024 — present	UniversalToolchain / Wist2 — Creator and Primary Developer Compiler SDK · diagnostics · verification infrastructure - Built a typed facade, structured diagnostics, and a versioned redacted trace for the public Wist API. - Added interpreter/CIL parity tests, negative contract tests, lifecycle checks, and architecture guard tests. - Added package-surface and clean-consumer smoke tests; the public baseline covers 75 .NET projects and 1,325 tests.
2026	PS-form Analyzer — Test Harness and Solver Verification Python generators · exact oracle · metamorphic testing - Built a Python harness with fixed and generated cases, strict stdout validation, and per-case timeouts. - Added an exact oracle for small domains plus randomized and metamorphic tests.

SELECTED PROJECTS

.NET · SDK · compiler/runtime tooling	UniversalToolchain / Wist2 A compiler/runtime framework with a public API, two execution backends, and dedicated testing infrastructure. - structured errors, versioned traces, and explanation APIs. - interpreter/CIL parity, structural verifiers, negative contracts, and state-isolation tests. Python · generative testing PS-form Verification Harness A generator and runner for a conservative memory-dependence analyzer. - Exact oracle for small domains and fixed regressions. - Randomized and metamorphic generation. C++23 · differential testing AdvancedAlgorithms Verification Infrastructure Validation for 12 reusable modules through reference algorithms, invariants, invalid-input cases, and large inputs. - Differential oracles for max flow, SCC, Dijkstra, HLD, and other modules. - ASan/UBSan, separate correctness/performance CTest labels, and API-contract checks.
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SKILLS

RECOGNITION

Programming olympiads Prize-winner of the HSE "Vysshaya Proba" Olympiad in competitive and industrial programming; prize-winner of the regional Russian Informatics Olympiad.
Compiler projects Two highest-tier diplomas in the MEPhI Junior competition and a highest-tier diploma in the Baltic Science and Engineering Competition.

EDUCATION

HSE University · Software Engineering · Faculty of Computer Science · bachelor's programme 09.03.04 · Moscow · Admitted; expected graduation in 2030.
